

Printed Organic Transistors for Ultra-Low-Cost RFID Applications

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Abstract—Printed electronics provides a potential pathway toward the realization of ultra-low-cost radio frequency identification (RFID) tags for item-level tracking of consumer goods. Here, we report on our progress in developing materials and processes for the realization of printed transistors for low-cost RFID applications. Using inkjet printing of novel conductors, dielectrics, and organic semiconductors, we have realized printed transistors with mobilities $>10^{-1}$ cm²/V-s. AC performance of these devices is adequate for 135-kHz RFID, and, with further optimization, 13.56-MHz RFID appears to be within reach. We review the performance of these devices, and discuss optimization strategies for achieving the ultimate performance goals requisite for realizing ultra-low-cost printed RFID.

Index Terms—Organic semiconductors, printed electronics, radio frequency identification (RFID).

I. INTRODUCTION

IN RECENT years, there has been significant interest in the development of radio frequency identification (RFID) tags for item-level tracking of individual consumer goods [1]. Such tags are expected to dramatically improve automation, inventory control, and checkout/purchasing operations, provided they are cheap enough to be widely deployed. For widespread deployment, a cost ceiling of 1–2 cents per tag has generally been indicated by most consumer product companies. Printed electronics is particularly promising for realizing such a low cost ceiling, since the cost of printed electronics is generally expected to be two to three orders of magnitude cheaper than silicon per unit area. We briefly review the basic cost assumptions driving the development of printed RFID tags, explore the various approaches being pursued, and describe our progress toward realizing the component technologies required to realize ultra-low-cost all-printed RFID.

Based on FCC regulations, RFID devices are generally being deployed in four main communication bands: 1) the low-frequency range up to 135 kHz, 2) a band at 13.56 MHz, 3) a band at 900 MHz, and 4) a band at 2.4 GHz [2]. Each band has its own advantages and disadvantages. The range of RFID tags is typically larger at higher frequencies for reasons relating to the antenna Q, and thus, most pallet-level RFID solutions make use of 900 MHz or 2.4 GHz [3]. Furthermore, since the

passive components (both inductors and capacitors) are smaller at these higher frequencies, total tag size can be reduced at these frequencies, reducing cost.

For item-level tracking, unfortunately, these frequencies do not work particularly well due to the presence of substantial amounts of metal (for example, soup cans, etc.), and RF-absorbing fluids (water, milk, etc.). Therefore, at this time, it appears that 13.56 MHz or lower is attractive from this perspective, since absorption is substantially reduced at lower frequencies. In general, therefore, the sweet spot for item-level RFID will likely exist at 13.56 MHz, though there is currently a push for developing 900-MHz solutions as well. Tag size at 135 kHz is generally too large, and likely too expensive for widespread item level tracking use, and operational use of the highest frequencies in metal-contaminated environments is correspondingly problematic.

II. CURRENT STATUS OF LOW-COST RFID DEVELOPMENT

Three main approaches are being pursued to realize ultra-low-cost RFID. In the conventional approach, low-cost silicon chips are used. The chip cost is miniscule, since typical tags (excluding the antenna) are less than 1 mm wide. Using various attachment technologies such as pick-and-place [4] and fluidic self assembly [5]), the tags are added to a paper or plastic strip containing the external antenna. Unfortunately, costs for this attachment operation do not appear to scale well at this time, and it seems unlikely that a silicon-based approach will realize sub-1-cent RFID in the foreseeable future. For cost reasons, silicon tags also generally operate in the higher frequency ranges, and are thus problematic in metal-contaminated environments, as discussed above.

A more aggressive approach is to make use of flexible electronics based on organic devices. Several groups are developing such technologies [6]–[8]. Using vacuum sublimation, good performance has been achieved using organic materials, in particularly pentacene. Carrier mobilities >10 cm²/V-s have been realized, with circuits operating at several MHz [7]. Unfortunately, all high-performance pentacene devices to date make use of slow deposition processes that reduce throughput to unacceptable levels. Correspondingly, the use of shadow-masking (required to achieve patterning) is economically uncertain at this time. If progress is made toward realizing high-performance, high-throughput technologies based on vacuum-sublimation and shadow-masking, these vacuum-sublimated organic thin film technologies (TFTs) may represent a disruptive path toward the realization of low-cost RFID.

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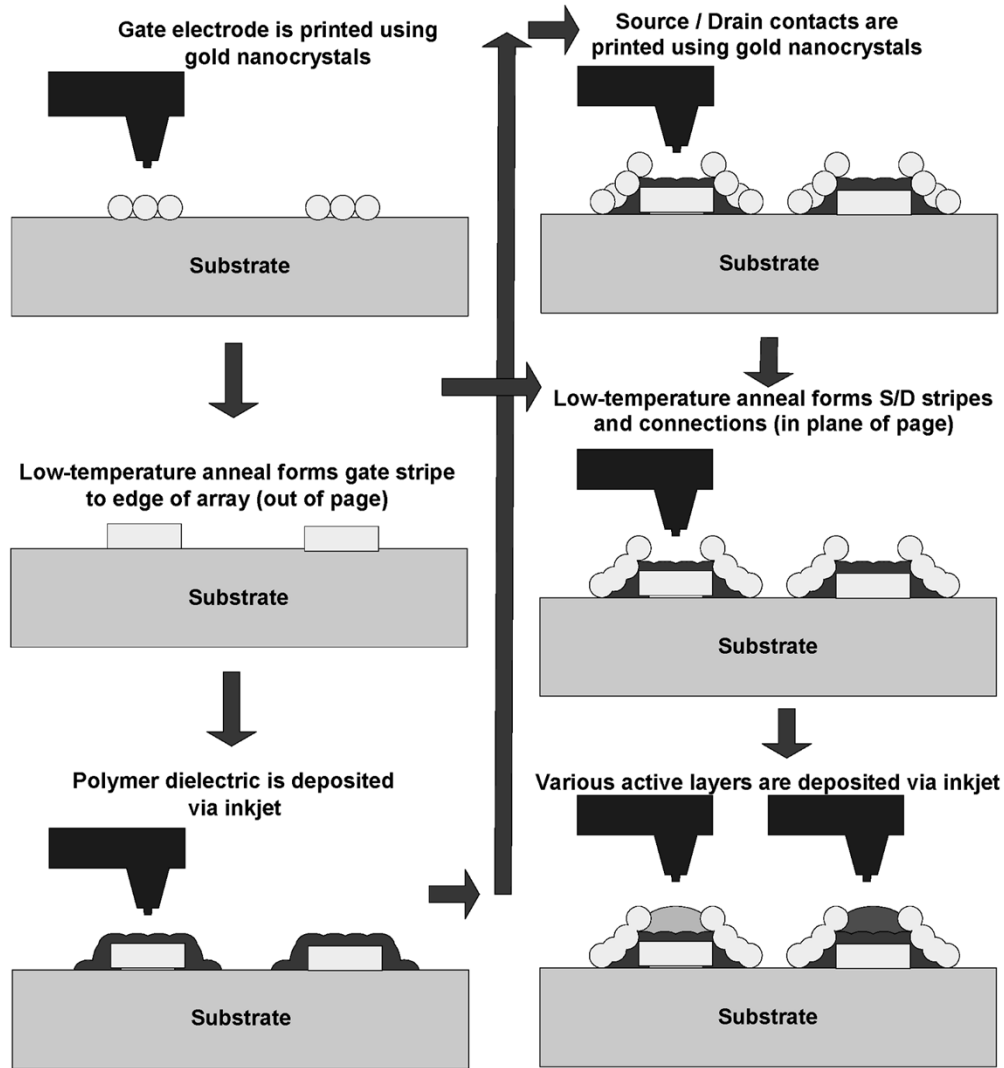


Fig. 1. Process-flow used to fabricate printed transistors. Various p-type and n-type printable semiconductors have been developed.

The final approach being considered for the realization of item-level RFID is all-printed organic electronics [8], [9]. In this process, an entirely additive printing process is used to realize printed devices and circuits, thus eliminating the need for lithography, vacuum processing, and subtractive process steps. Printing is also potentially an extremely high throughput technology. Thus, the costs associated with this technology are expected to be substantially lower than silicon-based techniques, particularly since the large area antenna is easily integrated using this process. To date, performance of printed devices is generally lower than the aforementioned vacuum-based technologies, though progress has been rapid and sustained. Here, we will describe the results obtained in our group that will hopefully allow performance to be improved to the point where all-printed item-level RFID will be technically and economically feasible.

III. PRINTED TRANSISTORS: STATUS AND OUTLOOK

To realize item-level RFID, both high-quality printed passive components (inductors and capacitors) and high-performance

all-printed transistors are required. We have previously reported on our printed passive component technologies [9]. Here, we will focus on printed active components, specifically, transistors. To realize printed transistors, we use the general process flow shown in Fig. 1 [10]. A metallic gate line is printed, followed by a polymer dielectric. Next, source and drain contacts are printed, followed by a printed semiconductor. The net result is the realization of a printed bottom-gate inverted staggered transistor structure. Using this structure, we typically obtain carrier mobilities of approximately $10^{-1} \text{ cm}^2/\text{V}\cdot\text{s}$ (Fig. 2), without any optimization of contacts or interfaces. AC performance of these devices is, unfortunately, inadequate for 13.56-MHz RFID applications, though it is likely adequate for 135-kHz RFID.

Based on limitations of our inkjetting technology, it is difficult to produce transistors with channel lengths less than $10 \text{ }\mu\text{m}$ without substantial overlap capacitance. This is apparent by looking at an optical micrograph of a sample printed device (Fig. 3). Current limitations of inkjet-printing limit printed gate length to $>30 \text{ }\mu\text{m}$ or larger. To achieve the short channel length necessary for high-speed, the source and drain and substantially overlap over the gate, resulting in large

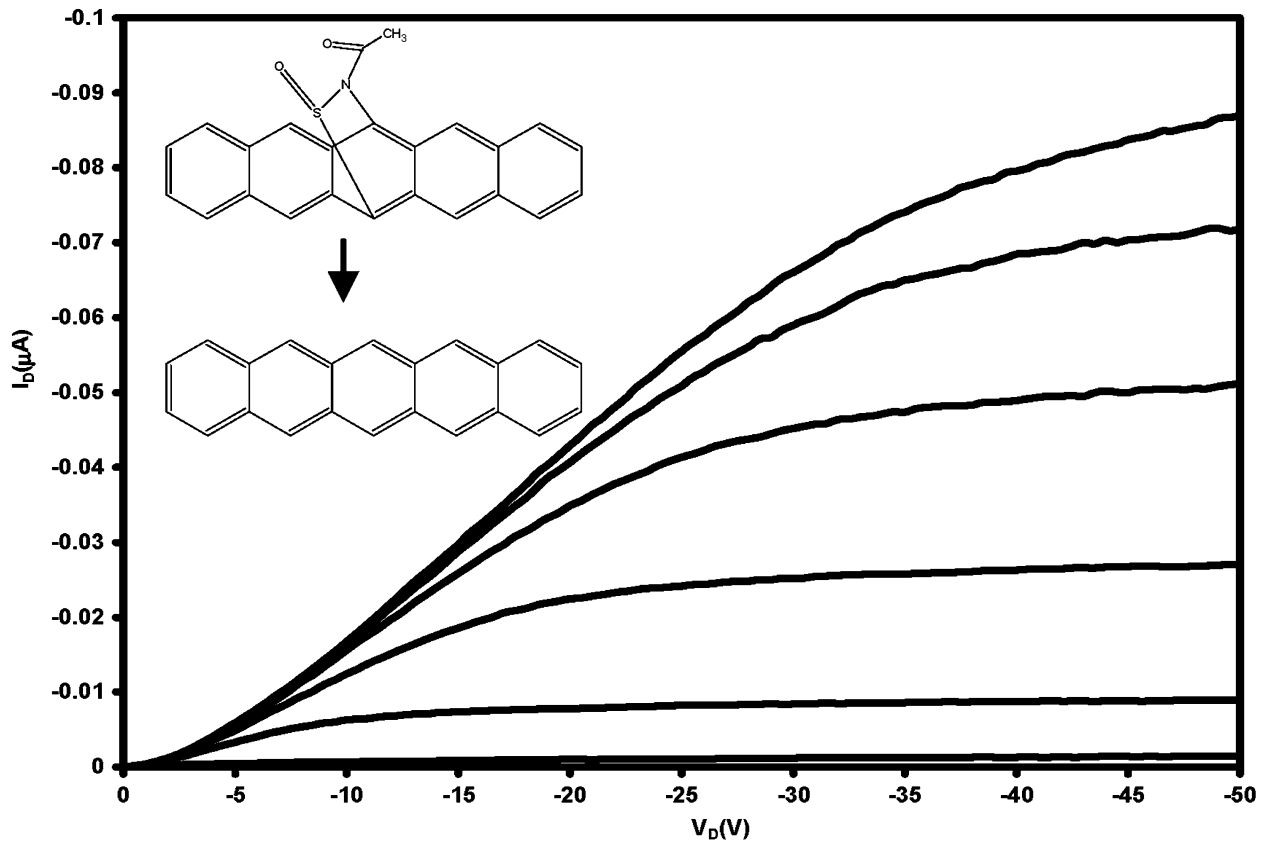


Fig. 2. Output characteristics ($V_{GS} = 0$ V to -60 V) of printed pentacene transistor using a pentacene precursor that converts to insoluble pentacene upon heating (inset).

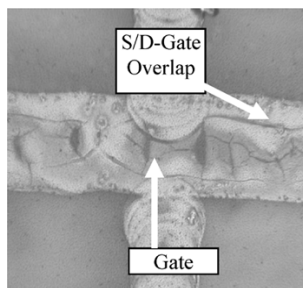


Fig. 3. Optical micrograph of inkjetted devices, showing large overlap capacitance issue (printed linewidth is ~ 120 μm).

capacitance. To understand the impact of this capacitance on performance, measurements were performed on devices designed with >100 μm overlap. As expected, the overlap capacitance causes substantial roll-off at ~ 10 kHz, which is inadequate for any RFID applications (Fig. 4). Fortunately, there is hope, however, since more precise printing technologies have been shown to reduce overlap to a ~ 5 μm . By accounting for this capacitance, achievable ac performance of these devices with reduced overlap has been estimated to be ~ 500 kHz, which is reasonable for 135-kHz RFID, but is inadequate for 13.56 MHz. Since the latter is the frequency of choice to realize low-cost tags, further improvement is required. Note that it isn't necessary for the tag circuitry to run at the carrier frequency; communication will likely be performed on a side-band using a subcarrier. However, to ensure the proper stability requisite for most communication strategies, this subcarrier will either

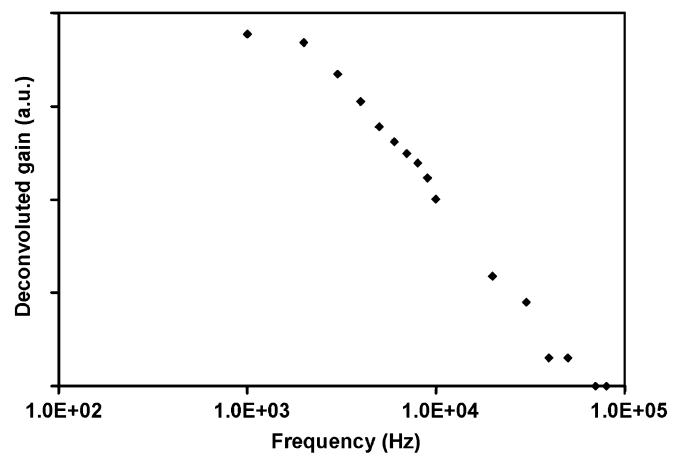


Fig. 4. AC performance of devices with known 100- μm overlap. Note the rolloff and 10 kHz, which extrapolates to a 500 kHz achievable performance.

have to be frequency-divided from the carrier or will have to be accurately phase-locked. Both solutions will likely require transistors with f_T significantly higher than we are currently able to achieve, or, for that matter have been achieved by other groups working on printed electronics.

Several different active layers may be used in the aforementioned process flow. We have demonstrated transistors with an active layer consisting of an annealed soluble pentacene precursor [11]. This material has been found to provide mobilities >0.3 $\text{cm}^2/\text{V}\cdot\text{s}$ in a printed, unoptimized structure. Other researchers have demonstrated mobilities approaching 1 $\text{cm}^2/\text{V}\cdot\text{s}$

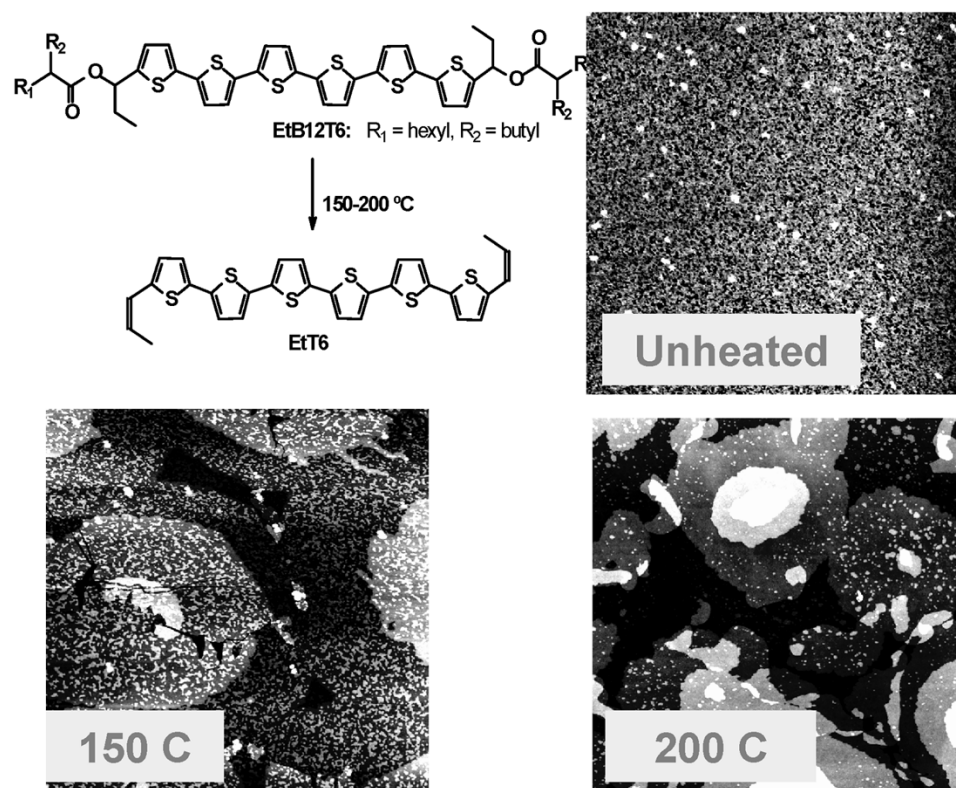


Fig. 5. Soluble oligothiophene EtB12T6 converts to the rigid molecule EtT6 with strong packing characteristics upon thermolysis (scan size 10 μm).

using this material in spin-cast films [12], so further improvement is certainly possible. The key to obtaining high-performance in organic materials lies in depositing channel films with excellent ordering; this ensures good pi-orbital overlap between adjacent molecules, allowing efficient hopping of carriers between molecules. To this end, we have initiated a program aimed at developing highly crystalline organic semiconductors formed by printing.

For this work, we have focused on oligothiophenes [13]. These deliver reasonably high mobilities, and are easily functionalized to produce printable precursors. This latter fact is important, since most small molecule organic semiconductors, including both pentacene and oligothiophene are insoluble and therefore not printable; they must be functionalized with solubilizing groups to ensure printability. In the approach used by various groups, including ours, the solubilizing groups are subsequently removed by heating to ensure the formation of highly crystalline films.

Most recently, we have demonstrated a novel oligothiophene functionalized with thermally labile solubilizing groups (Fig. 5). After printing, the films are heated, removing the same. During this transition, the molecules show high surface mobility and reorganize to form crystalline structures. Due to the high crystallinity, mobilities of $\sim 10^{-1} \text{ cm}^2/\text{V}\cdot\text{s}$ have been obtained in unoptimized structures, suggesting that substantial performance improvement is possible using this approach [14].

Aside from improving mobility, off-state behavior and stability are also equally important concerns for RFID. Since RF tags are typically power-constrained, good off-state behavior is crucial. Stability is important for good analog circuit operation.

We have exploited the high surface-mobility of our oligothiophene derivatives to demonstrate devices with excellent off-state behavior and stability without sacrificing on-state behavior.

In all field-effect transistors (FETs), on-state conduction typically happens within the first few nanometers of channel material near the gate interface. The remaining material contributes purely to off-state leakage. This has resulted in great interest in ultra-thin-body devices, which eliminate these extra leakage paths. We have achieved a similar structure via printing of our oligothiophene materials. By using appropriate surface chemistries and ink dilutions, we can exploit the high-surface mobility of our oligothiophene derivative to ensure that the printed channel droplet reorganizes to form a device with a channel consisting entirely of a single monolayer [15]. Since this molecule is only 3-nm-tall, this results in the formation of an ultra-thin body device with excellent electrostatics (Fig. 6). Also, serendipitously, it appears that by eliminating the thick film structure, we are able to greatly reduce the trap density in the active layer, resulting in devices with substantially reduced hysteresis and improved sub-threshold slope (Fig. 7).

The final two issues that remain to be addressed are the high operating voltages of these devices, and the absence of NMOS materials. Low operating voltage is required to provide adequate operating range; indeed, the 20–30 V operation in most of the devices above is entirely unacceptable for RFID—sub-5-V operation is clearly required. Fortunately, a solution in this regard is apparent based on the research of several groups. All the devices above make use of relatively thick dielectrics (100-nm EOT or more). To reduce operating voltage, the EOT of these devices may be reduced. Unfortunately, the reliable printing of

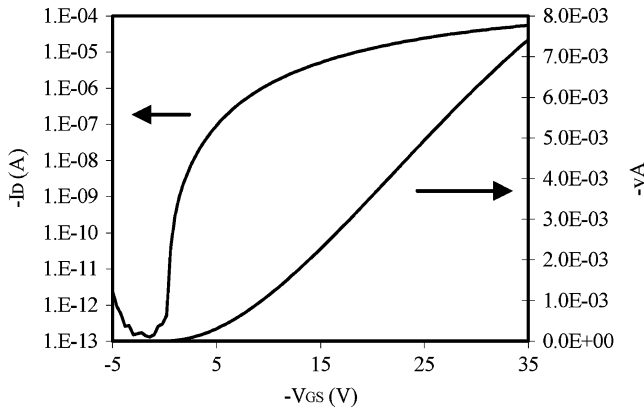


Fig. 6. ID-VG characteristic for printed monolayer device ($W/L = 250/3$, $V_D = -35$ V), showing excellent off-state characteristics due to near-ideal electrostatic configuration.

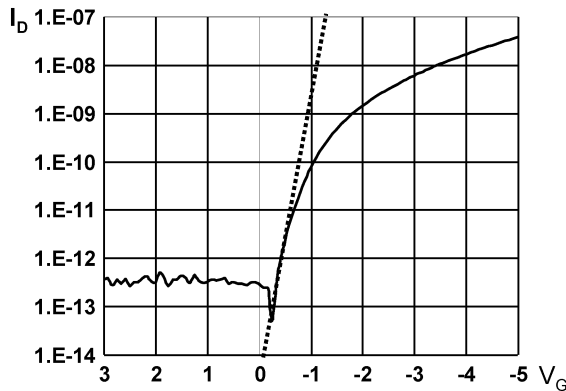


Fig. 7. Subthreshold characteristic of a monolayer device showing excellent subthreshold slope <200 mV/dec, which is impressive considering the 100-nm EOT dielectric. In comparison, most printed devices have swings >1 V/dec.

thinner layers is difficult; however, alternative strategies that look promising have been demonstrated.

In a first strategy, a self-assembled monolayer has been used to demonstrate 2-V operation and 100-mV/dec swing [16]. This technique may be printing-compatible. In a second strategy, high-K dielectrics may be used [17]. We have demonstrated the use of soluble high-K dielectric precursors to demonstrate devices with sub-4-V operation (Fig. 8) [9]. While substantial work remains to be done to reduce low-field leakage, etc., the promise of this technique is clear.

The dielectric is an important concern in all field-effect devices, and relatively little effort has been dedicated to this in printed electronics. Aside from the performance implications above, recent results indicate that the dielectric interface also contributes to degraded stability; electrical stressing of printed devices results in degradation of drive-current without a degradation of mobility (Fig. 9). This indicates that the primary degradation mechanism is a threshold voltage shift, likely taking place at/near the dielectric interface.

All the results above focus on p-channel metal-oxide semiconductor (PMOS). The absence of an adequate n-type MOS (NMOS) material is a barrier to realization of complementary MOS (CMOS), with its associated design simplicity and

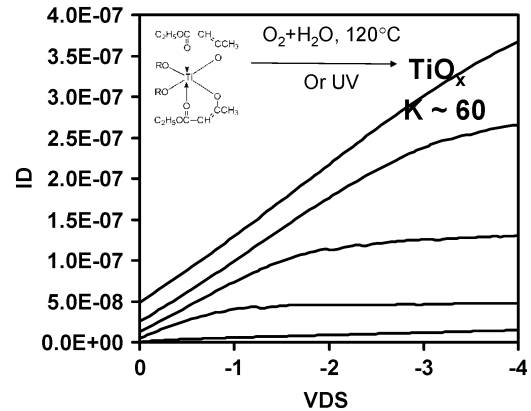


Fig. 8. Output characteristics of a thiophene-based TFT with a high-k dielectric. The large low-leakage is evident, and is caused by the formation of a porous, leaky TiO_x dielectric from the Ti-precursor (inset).

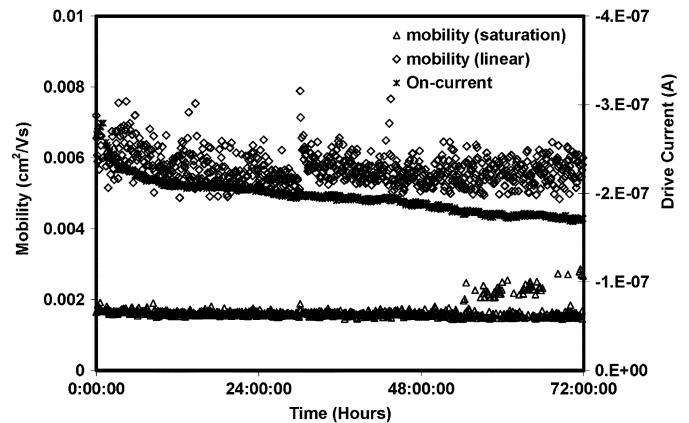


Fig. 9. Degradation characteristics of printed FETs showing mobility stability versus drive current degradation, measured in an inert environment. This degradation is due to V_T instability.

low-power benefits. All organic NMOS materials to date have poor performance ($<10^{-2}$ $\text{cm}^2/\text{V}\cdot\text{s}$ mobility) and stability [18]. Recently, we have demonstrated an alternate solution to this problem. Using printed Zinc oxide nanoparticles, we have demonstrated stable NMOS devices with mobilities of >0.1 $\text{cm}^2/\text{V}\cdot\text{s}$ [19]. While the results are early, they attest to the potential of a hybrid approach to realizing printed electronics.

IV. CONCLUSION

Printed electronics provides a promising potential pathway toward the realization of ultra-low-cost RFID tags for item-level tracking of consumer goods. Using printing, we have realized printed transistors formed entirely using printed films including various novel organic semiconductors, dielectrics, and nanoparticle-based conductors. While the performance of these devices is currently inadequate for operation at the frequency of choice, i.e., 13.56 MHz, improvement has been steady, and further improvements in knowledge along with the associated technology optimization should ultimately enable the realization of all-printed RFID tags, ushering in an era of enhanced inventory efficiency and consumer convenience.

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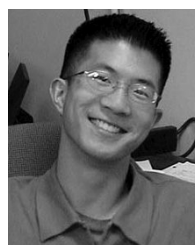


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